MODULE 21 · DOMAIN 5 - INCIDENT AND EVENT RESPONSE

Event-Driven Operations

EventBridge rules, patterns and buses, CloudTrail and AWS Health events, cross-account delivery, reliability, Scheduler, Pipes and Step Functions.

DOMAIN 5 of 6 EXAM WEIGHT 14% MODULE 21 of 34

admin@sthithapragna.com

Operations becomes event-driven when every change announces itself, and something is already listening.

Event, bus, rule, target

01
Event
A JSON record: source, detail-type, detail.

02
Event bus
Receives events: default, custom or partner.

03
Rule
An event pattern selects the events it wants.

04
Target
Up to five per rule: Lambda, SQS, runbooks…

05
Outcome
Notify, remediate, record, or start a workflow.

ON THE EXAM

When a rule 'never fires', check each link: was the event produced, on this bus, and does the pattern match exactly?

Where events come from

SOURCE	EXAMPLES
AWS service state changes	Instance stopped, deployment failed, pipeline succeeded
CloudTrail API calls	AuthorizeSecurityGroupIngress, DeleteBucket, CreateUser
AWS Health	Scheduled retirement, maintenance, service issues
CloudWatch alarms and Config	Alarm state change, rule compliance change
Security services	GuardDuty findings, Security Hub findings, Inspector findings
Your applications and partners	PutEvents from your code; SaaS partner events

ON THE EXAM

Many services emit their own state-change events directly. API calls come through CloudTrail.

An event pattern

```
1  {
2    "source": ["aws.ec2"],
3    "detail-type": ["AWS API Call via CloudTrail"],
4    "detail": {
5      "eventName": ["AuthorizeSecurityGroupIngress"],
6      "requestParameters": { "ipPermissions": { "items": {
7        "ipRanges": { "items": { "cidrIp": ["0.0.0.0/0"] } } } } },
8      "userIdentity": { "arn": [{ "anything-but": {
9        "prefix": "arn:aws:iam::111122223333:role/NetworkAdmin" } }] }
10   }
11 }
```

5, 7 Someone opened a rule to the whole internet.

8-9 Unless it was the network admin role: content filtering.

ON THE EXAM

Every field listed must match; values in an array are alternatives. Matching is exact and case-sensitive.

CloudTrail events: the trap

- 'AWS API Call via CloudTrail' events need a **CloudTrail trail with logging enabled**.
- Write management events match rules in the normal ENABLED state.
- Read-only management events need the rule state **ENABLED_WITH_ALL_CLOUDTRAIL_MANAGEMENT_EVENTS**.
- S3 object-level calls are data events: the trail must log data events for them.

> ON THE EXAM
>
> A rule on API calls never fires in a new account: there is no trail. Event history alone is not enough.

Three kinds of event bus

BUS	RECEIVES	TYPICAL USE
Default	Events from AWS services in the account	Operations automation
Custom	Your applications' events, and other accounts'	Application events, central buses
Partner	Events from SaaS partners	Integrations such as ticketing tools

ON THE EXAM

A bus's resource policy decides which other accounts may put events on it, including whole organisations.

Crossing accounts: direct target, or bus to bus?

DIRECT CROSS-ACCOUNT TARGET

January 2025

- Rule targets a queue, function or topic in another account
- Targets with resource-based policies only
- No intermediate bus to run
- Receiving team just grants access

BUS TO BUS

The established route

- Rule forwards to a bus in the other account
- A rule there picks the final target
- Works for any target, including pipelines
- Receiving account controls routing

THE TIEBREAKER

Target has a resource policy (SQS, Lambda, SNS, Kinesis, API Gateway): direct. Pipelines and workflows: bus to bus.

Common operational targets

TARGET	USE IT TO
SNS topic	Notify people and systems, with fan-out
SQS queue	Buffer events for a worker, absorbing bursts
Lambda function	Run a short custom action
Step Functions	Run a multi-step workflow with waits and approvals
Systems Manager	Run an Automation runbook or Run Command
CodePipeline, CodeBuild, ECS task	Start a release, a build, or a batch job

ON THE EXAM

An input transformer reshapes the event before it reaches the target, for example into a readable message.

Making delivery reliable

- A **retry policy** retries failed deliveries, by default for up to 24 hours and 185 attempts.
- A **dead-letter queue** on the target keeps events that still could not be delivered.
- An **archive** keeps events for a chosen period; **replay** sends a time window back through the bus.
- A rule whose target triggers the same rule again creates an infinite loop, and a large bill.

> ON THE EXAM
>
> Reprocess events after fixing a consumer bug: archive and replay. Keep undeliverable events: a target DLQ.

EventBridge Scheduler, or a scheduled rule?

EVENTBRIDGE SCHEDULER

The newer scheduler

- Time zones, with daylight saving handled
- One-time and recurring schedules
- Flexible windows to spread load
- Targets thousands of AWS API actions

SCHEDULED RULE

The older way

- Cron or rate expressions in UTC
- Recurring only
- Lives on an event bus
- Fewer target types

THE TIEBREAKER

'At 18:00 London time, all year': Scheduler with a time zone. A UTC cron rule drifts by an hour twice a year.

EventBridge, SNS, or Pipes?

EVENTBRIDGE PIPES

Point to point, enriched

- One source to one target
- Sources: SQS, Kinesis, DynamoDB Streams, Kafka
- Filter and enrich on the way
- Replaces glue Lambda functions

EVENTBRIDGE BUS OR SNS

Many to many

- Bus: rich content-based routing
- SNS: high-throughput fan-out, filter policies
- Both push to many subscribers
- SNS also reaches email, SMS, mobile

THE TIEBREAKER

Poll a stream, filter, enrich, deliver to one target, without code: Pipes. Route many events to many consumers: a bus.

Step Functions for operational workflows

- Wait, retry with backoff, branch on conditions, run steps in parallel, and catch errors.
- Pause for a person with a **task token**, then continue when they respond.
- Call over 200 AWS services directly through SDK integrations, often with no Lambda.
- Standard workflows run up to a year; Express workflows suit high-volume, short runs.

> ON THE EXAM
>
> A response with several steps, waits, approvals and retries: Step Functions, not one long Lambda function.

An automated security response

01
Finding
GuardDuty reports a compromised instance.

02
Rule
Matches high-severity findings only.

03
Isolate
Swap to a no-access security group.

04
Preserve
Snapshot the volumes for forensics.

05
Approve
A person decides whether to terminate.

06
Record
Notify, and open an OpsItem.

ON THE EXAM

The event starts it; the workflow carries it through. Detection to containment in seconds, with a person for the irreversible step.

AWS Health across an organisation

- AWS Health events arrive in each account's default bus as source aws.health.
- The **organisational view** gathers Health events from every account, with a delegated administrator.
- Rules act on scheduled changes before their deadlines: retirements, maintenance, certificate expiry.
- Forward Health events to a central operations bus for one team to handle.

ON THE EXAM

Scheduled retirement in any of 80 accounts must be handled automatically: Health events, forwarded centrally, with a rule and runbook.

> In a newly created account, a security engineer creates an EventBridge rule on the default bus that matches 'AWS API Call via CloudTrail' events for AuthorizeSecurityGroupIngress, targeting an SNS topic.
>
> Security groups are changed several times a day, but the rule never fires. The account has no CloudTrail trails.

What should the security engineer do?

A Remove detail fields from the pattern so that it matches more broadly

B Add a second target to the rule

C Create a CloudTrail trail with logging enabled for management events

D Set the rule state to ENABLED_WITH_ALL_CLOUDTRAIL_MANAGEMENT_EVENTS

◆ PAUSE AND DECIDE

What should the security engineer do?

A **Broader pattern**

No events are arriving at all. A broader pattern still matches nothing.

B **Second target**

The rule never fires, so no target is ever invoked.

C **Create a trail**

API call events need a trail with logging enabled. Event history alone does not deliver them.

D **All management events state**

Needed for read-only calls. This is a write call, and there is still no trail.

THE SIGNAL

API-call rule silent in a new account, 'no trails': the events are never produced.

> A security team wants every S3 bucket policy change in a workload account delivered to an SQS queue that it owns in its security account.
>
> The security team does not want to manage any EventBridge resources, and the company wants the fewest moving parts.

What should the DevOps engineer set up?

A. A rule in the workload account that targets the security account's queue directly, with a queue policy allowing it

B. A rule forwarding events to a custom bus in the security account, plus a rule there targeting the queue

C. A Lambda function in the workload account that sends each event to the queue

D. An SNS topic in the workload account, with the queue subscribed across accounts

◆ PAUSE AND DECIDE

What should the DevOps engineer set up?

A Direct cross-account target

Since January 2025: the rule delivers straight to the queue; the security team only writes a queue policy.

B Bus to bus

Works, but puts a bus and a rule in the security account, which the team wanted to avoid.

C Forwarding Lambda

Custom code doing what EventBridge now does natively.

D SNS in between

An extra hop and service for no benefit.

THE SIGNAL

Target in another account with a resource policy, and no EventBridge for the receivers: direct delivery.

> An EventBridge rule sends order events to a partner's HTTP endpoint through an API destination. During a partner outage lasting several days, events were dropped once retries ran out.
>
> The company wants no event to be lost, and wants to send the missed events once the partner recovers.

What should the DevOps engineer configure?

A. A higher invocation rate limit on the API destination

B. A second rule with the same pattern and target

C. An input transformer on the target

D. A dead-letter queue on the target, keeping undelivered events for redelivery

◆ PAUSE AND DECIDE

What should the DevOps engineer configure?

A **Higher rate limit**

Sends faster to an endpoint that is down. Nothing is kept.

B **Duplicate rule**

Both copies fail in the same way, and duplicates follow when the partner recovers.

C **Input transformer**

Reshapes events. Does nothing for delivery failures.

D **Target dead-letter queue**

After retries are exhausted, events land in the queue instead of vanishing, ready to resend.

THE SIGNAL

Retries exhausted, events dropped: add a target DLQ. Retries cover minutes to a day; a DLQ covers longer.

> Development instances must stop at 18:00 London time every weekday and start at 07:30. A rule written in UTC cron has run an hour late for half of every year.
>
> The team also wants one-off stops before public holidays, with the least operational overhead.

What should the DevOps engineer use?

A Two scheduled EventBridge rules, edited by hand when the clocks change

B EventBridge Scheduler schedules in the Europe/London time zone, including one-time schedules, calling the EC2 API

C A cron job on a small always-on instance

D A Lambda function that runs every minute and checks the time

◆ PAUSE AND DECIDE

What should the DevOps engineer use?

A **Rules edited twice a year**
The same UTC problem, fixed by hand, forever.

B **Scheduler with a time zone**
Daylight saving handled, one-time schedules supported, and EC2 called directly.

C **Cron on an instance**
A server to run, patch and keep alive, to tell the time.

D **Lambda every minute**
Over 40,000 invocations a month to do the work of four schedules.

THE SIGNAL

Local time with daylight saving, plus one-off runs: EventBridge Scheduler.

> Order events flow through a custom event bus to several consumers. Last month a consumer bug mishandled six hours of events, and they could not be reprocessed.
>
> The company wants to be able to reprocess any events from the past seven days after fixing a future consumer bug.

What should the DevOps engineer set up?

A An archive on the custom bus with seven days' retention, and replay when needed

B A dead-letter queue on each target

C A CloudTrail trail with data events

D Log every event to CloudWatch Logs and write a script to republish them

◆ PAUSE AND DECIDE

What should the DevOps engineer set up?

A Archive and replay

Keeps events for seven days and replays any time window through the bus after the fix.

B Target DLQs

Catch failed deliveries. These events were delivered, and then mishandled.

C CloudTrail data events

Records API calls, not the application's event stream.

D Logs and a republish script

Rebuilds archive and replay by hand.

THE SIGNAL

Delivered but mishandled: DLQs cannot help. Archive and replay can.

> On a high-severity GuardDuty finding for an EC2 instance, the security team wants to isolate the instance, snapshot its volumes, wait for an engineer to approve termination, and then terminate or release it.
>
> Each step needs retries, and the approval may take hours.

What should the DevOps engineer build?

A. A single Lambda function that performs every step and waits for approval

B. An SNS email to the team asking them to carry out the steps

C. An EventBridge rule on high-severity findings that starts a Step Functions workflow with a task-token approval step

D. Several EventBridge rules, each triggering the next step

◆ PAUSE AND DECIDE

What should the DevOps engineer build?

A One Lambda function

Lambda stops after 15 minutes. An approval taking hours cannot fit.

B Email the team

Manual, slow, and containment waits for someone to read it.

C Rule and Step Functions workflow

Retries per step, a task token for the approval, and waits of hours or days.

D Chained rules

No state, no approval step, and hard to follow or retry.

THE SIGNAL

Several steps, retries, a human approval that takes hours: an event starts a Step Functions workflow.

Module 21 in five lines

- Event, bus, rule, target. Patterns match exactly; arrays are alternatives.
- API-call events need a trail; read-only calls need the all-management-events rule state.
- Cross-account: direct to resource-policy targets since January 2025; bus to bus for the rest.
- Retries and target DLQs for failed delivery; archive and replay for mishandled events.
- Scheduler for time zones and one-offs; Pipes for enriched point to point; Step Functions for steps.

ON THE EXAM

Next, module 22: automated remediation with AWS Config, runbooks and OpsCenter.